

# Neonatal Jaundice

## Contents

|                                                 |          |
|-------------------------------------------------|----------|
| <b>Neonatal Jaundice</b>                        | <b>1</b> |
| Definition . . . . .                            | 1        |
| Bilirubin Metabolism (why it happens) . . . . . | 1        |
| Classification . . . . .                        | 1        |
| Causes by timing . . . . .                      | 1        |
| Clinical assessment . . . . .                   | 2        |
| Complication: Kernicterus . . . . .             | 2        |
| Management . . . . .                            | 2        |
| Key takeaways . . . . .                         | 2        |

## Neonatal Jaundice

### Definition

Neonatal jaundice is the yellowish discoloration of the skin, sclera, and mucous membranes in a newborn, caused by raised levels of bilirubin in the blood (hyperbilirubinaemia). It becomes clinically visible when serum bilirubin exceeds roughly 85 micromol/L (5 mg/dL).

### Bilirubin Metabolism (why it happens)

Bilirubin is the breakdown product of haem, mostly from old red blood cells. In the newborn:

- Red cell lifespan is shorter (70–90 days vs 120 in adults), so more haem is broken down.
- The liver enzyme **UDP-glucuronosyltransferase (UGT)**, which conjugates bilirubin to make it water-soluble, is immature at birth.
- The gut has increased **enterohepatic circulation**, reabsorbing bilirubin back into the blood.

Together these make newborns prone to a temporary rise in unconjugated (indirect) bilirubin.

### Classification

1. **Physiological jaundice** — appears after 24 hours, peaks day 3–5, resolves by day 14. Benign, unconjugated.
2. **Pathological jaundice** — any of: appears within the first 24 hours, bilirubin rising fast ( $>85$  micromol/L/day), conjugated fraction high, or persisting beyond 14 days. Always needs a cause.

### Causes by timing

- **< 24 hours:** always pathological — haemolysis (Rhesus or ABO incompatibility, G6PD deficiency), congenital infection.

- **24 hours – 14 days:** physiological, breastfeeding jaundice, sepsis, polycythaemia, bruising/cephalhaematoma.
- **> 14 days (prolonged):** breast-milk jaundice, hypothyroidism, biliary atresia (conjugated — a surgical emergency), UTI, galactosaemia.

## Clinical assessment

Assess in good natural light. Jaundice progresses **cephalocaudally** (face first, then trunk, then limbs — Kramer’s rule), so palms and soles being yellow indicates severe jaundice. Always measure bilirubin objectively — transcutaneous or serum — rather than relying on the eye.

## Complication: Kernicterus

Unconjugated bilirubin is fat-soluble and can cross the immature blood–brain barrier, depositing in the basal ganglia. This causes **acute bilirubin encephalopathy** (lethargy, poor feeding, high-pitched cry, hypertonia, opisthotonus) which can progress to **kernicterus** — permanent choreoathetoid cerebral palsy, sensorineural hearing loss, and gaze palsy. This is the outcome all treatment aims to prevent.

## Management

- **Phototherapy:** blue light (450–460 nm) converts bilirubin to water-soluble isomers excreted without conjugation. First-line, guided by age-specific threshold charts.
- **Exchange transfusion:** for dangerously high or rapidly rising levels, or signs of encephalopathy.
- **Treat the cause:** antibiotics for sepsis, surgery for biliary atresia, etc.
- **IV immunoglobulin** in isoimmune haemolytic disease.

## Key takeaways

Physiological jaundice is common and benign; the job is to spot the pathological patterns (first 24 hours, prolonged, or conjugated) and to prevent kernicterus.